Content

Learning objectives and content

Objectives

After completing this course, the students should be able to:

1. Explain the importance of software evolution and describe how it manifests in software systems.
2. List important challenges associated with software evolution.
3. Discuss methods and tools addressing these challenges, their advantages, and disadvantages.
4. Apply the methods and tools to existing software systems and interpret the results obtained in a scientifically responsible way.
5. Analyze and describe the evolution of a real-life software system.

Content

Nowadays change is often considered as the only constant factor in software development. Successful software systems are, therefore, those systems that can adapt to the ever-changing requirements of the environment. One can, thus, compare this process of adaptation to Darwin's survival of the fittest principle. The problem is, however, that not much is known about the evolution of software systems. This course will explore issues related to software evolution: how it manifests, why it is difficult, how we can cope with this difficulty, and what we can learn from the past. Specifically, the course will look at advanced tools and techniques proposed by the research community to understand, ease and automate software evolution. We will discuss requirements evolution, architecture reconstruction, code cloning and differencing, mining software repositories, software metrics, software ecosystems, the evolution of tests, refactoring, and reengineering, among other topics. In this course, you will have a chance not only to learn about methods and tools of software evolution, but also to apply them to assess software the evolution of an existing software system.

2026

Software Evolution (2IMP25)

Practical information

Course module

2IMP25

Course type

Graduate School

Credits (ECTS)

5 EC

Category

Language of instruction

English

Offered by

Mathematics and Computer Science - Computer Science domain

Course enrolment

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block GS3
  Timeslot(s)
  E1 - Tu 5-6, Th 1-2

  Enrolment period
  15 November 2026 until 3 January 2027 23:59

  Unenrolment
  until 21 March 2027 23:59

Entrance requirements

Assumed previous knowledge
Mandatory: An undergraduate software engineering course including an up-to-date knowledge of modern software engineering techniques and processes. We do not expect you to know the most recent frameworks or the exact differences between different versions of Java; however, you should be able to read and understand source code and be able to contribute to a large open-source project using Git-based systems.

Previous knowledge can be gained by
Previous knowledge can be gained by Recommended: None of the following is mandatory but it can help when working on the assignments. In particular, if you realize that you miss some of the recommended skills you should consider teaming up with a student who has these skills: • Familiarity with statistical techniques (e.g., you should understand what R2 means) • Familiarity with academic writing in English (SFC640)

Instructional modes

Instructional modes

- Laboratory
  Remark

- Lecture
  Remark

Tests

Tests

- Homework assignment
  Interim examination

  Test weight
  60

  Minimum grade
  5.0

- Written examination
  Final examination

  Test weight
  40

Minimum grade
5.0

Test enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Homework assignment

- Block GS3
  Enrolment period
  15 November 2026 until 21 March 2027 23:59
- Block GS4
  Enrolment period
  15 November 2026 until 6 June 2027 23:59

Written examination

- Block GS3
  Enrolment period
  15 November 2026 until 21 March 2027 23:59

  Unenrolment
  until 21 March 2027

  Test date
  6 April 2027
- Block GS4
  Enrolment period
  15 November 2026 until 6 June 2027 23:59

  Unenrolment
  until 6 June 2027

  Test date
  29 June 2027

Lecturer(s)

Contactperson for the course

- L.M. Ochoa Venegas, MSc

Responsible lecturer

- L.M. Ochoa Venegas, MSc

Subject matter expert

- L.M. Ochoa Venegas, MSc

Materials

Do you want to know more?

L.M. Ochoa Venegas, MSc can tell you more about this course.
Send an e-mail to l.m.ochoa.venegas@tue.nl